

Name (Last, First):

NEU ID:

Assignment 4

METCS544A2_S2024

Please type or handwrite your answers clearly and concisely. Show all work - credit will not be given for numerical solutions that appear without explanation. **[Total 63=3+60 pts]**

Grading Rubric

Each question is worth 3 points and will be graded as follows:

3 points: Correct answer with work shown

2 points: Incorrect answer but attempt shows some understanding (work shown)

1 point: Incorrect answer but an attempt was made (work shown)

0 points: Left blank or made little to no effort/work not shown

Reflective Journal [3 pts]

(link to your live google doc)

I. Collecting Data: Sampling (9 x 3 = 27 pts)

Your local movie theater has put in a lot of work to make their customer's movie going experience great. They would like to take a random sample of their recent customers and email out a survey asking about their experience.

1) For the following situations, identify the bias and explain how they would introduce bias into the results.

a) You send out an email to the first 10 customers that buy tickets to a movie.

Answer:

b) Of the 10 movies that are playing in a single day, you randomly select one movie and email the survey to everyone who attended that movie.

Answer:

c) You set up a sign as the patrons exit the theater and tell patrons to email support@movies.com if they would like to share their movie going experience.

Answer:

d) You have the manager of the movie theater waiting outside the movie theater and interviewing every 10th person who walks out about their movie going experience.

Answer:

You obtain a list of the following data in a month at the movie theater: Order ID, Date, Customer Name, Customer Email, Movie Title, Show Time, Number of Tickets Purchased.

2) You would like to email out a survey to a sample of customers. For the following sampling methods, describe how you would use the information above to select a sample.

a) Simple Random Sample

Answer:

b) Stratified Random Sample

Answer:

c) Cluster Random Sample

Answer:

d) Systematic Random Sample

Answer:

e) A Census

Answer:

II. Collecting Data: Selecting Random Samples (11 x 3 = 33 pts)

1) The local university is interested in how many of its students plan on using some of the new facilities it installed over the summer. There are 3000 students who attend the university and they want to send out a questionnaire to 100 of them. They decide to take an SRS from an alphabetized list of students.

(a) Describe how you will select the sample.

Answer:

(b) Below is a sample of the table of random digits. Using the table and then method described above, what are the first four numbers that will be part of your sample?

23328 99330 01231 42492 73831 02911 01524 32932 34334 74280 29357 29301

Answer:

2) Your school has a pep rally in the gymnasium. The following is a map of the gym, where each box represents a seat that a student sits in.

1	2	3	4	5	G y m F i o r	301	302	303	304	305
...	...	...	...	...		...	...	...	...	...
146	147	148	149	150		446	447	448	449	450
151	153	154	155	156		451	452	453	454	455
...	...	...	...	...		...	...	...	...	...
296	297	298	299	300		596	597	598	599	600

Freshman: Seats 1 – 150

Sophomores: Seats 151 – 300

Juniors: Seats 301 – 450

Seniors: Seats 451 – 600

Describe how you would use a random number generator to select 40 students to complete a survey with each of the following sampling methods. **NOTE: I want you to describe it, not actually do it!*

(a) Simple Random Sample

Answer:

(b) Stratified Random Sample

Answer:

(c) Cluster Sample

Answer:

3) [A study was done](#) to determine if eating chocolate during pregnancy has an effect on infant temperament at the age of 6 months. The study asked about 300 pregnant women to report their chocolate consumption (Never/Seldom, Weekly, and Daily). Six months after they had given birth, the researchers then asked the mothers to report their child's overall temperament (activity, smiling, fear, etc.). They determined "Maternal prenatal consumption of chocolate was associated with the infant temperament in this study. Mothers who reported consuming chocolate daily rated their infants at 6 months as more positively reactive and active". Was this an observational study or an experiment? Explain.

Answer:

4) [A large study](#) in the UK was done to determine if there was a link between cell phone usage and brain cancer. The study set out to determine if using a cell phone more, increased the likelihood of getting brain cancer. The study took about 800,000 women and asked them to report their cell phone use in 1999, in 2005, and again in 2009. In 2009, they recorded the proportion of women who had developed brain cancer. Their conclusion was "In this large prospective study, mobile phone use was not associated with increased incidence of glioma, meningioma or non-CNS cancers." Is this an observational study or an experiment? Justify your answer.

Answer:

5) McGraw Hill would like to test synchronous vs asynchronous learning for a new Algebra I classroom. They give a test to a group of 200 students who are brand new to the Algebra I classroom. They then assign half of the students to study Algebra I synchronously (with a classroom and traditional teacher) and half of the students to study Algebra I asynchronously (online and at their own pace). At the end of the year, students were given the same test and the difference between the first test and the second test were compared. Is this an observational study or an experiment? Justify your answer.

Answer:



THE END